TM Forum Guidebook

GB1022 ODA Functional Architecture Guidebook

GB1022

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Executive Summary

The fifth generation of mobile technology and onward, seize growth opportunities that drives continuous management and operations transformation. This demands a “fifth generation” business that is primarily digital-first. The result is addressing business transformation in a holistic way, a way encompassing strategy transformation and continuous innovation; customer centricity; digital business operations, responsiveness and agility; and the ability to engage in multi-sided and platform-based business models.

Supporting systems that have in the past helped service provides in technology business management and operations, typically referred to as Business and Operations Support Systems (BSS/OSS) hold the key to manage and monetize new opportunities at scale. The ability to align and transformation these systems has mandated refactoring of the typical two-view monolithic view, into a flat, non-layered management and operations supporting systems.

This guidebook presents the standards for refactoring typical BSS/OSS into a digital-native architecture that facilitates simplification of business operations. As service providers evolve and refactor technology systems, CTIOs must address Customer Experience needs, Revenue Management needs, Total Cost of Ownership (TCO) and effective Operational Efficiency. These are the agility and responsiveness goals that mark the success of businesses in a digital-native world. The new generation of systems needed, embrace advancement in prior architectural models, like Service Oriented Architectures (SOA), event-based Microservice with new development and operations models that rely on Open source, Virtualization and Software-enabled management. The result is the ability to scale technologies to meet business needs, and to ultimately achieve business goals.

The business potential leading to defining a new management and operations software stack has been explored in many TM Forum research reports, and member collaboration projects. These initiatives have altogether explored the opportunities a service-oriented operation, solving the problems of a communications service provider, and attaining utilizing the best of technologies and capabilities from both the IT and CT worlds. This combination is refining existing and new business and addressing new growth opportunities.

The Open Digital Architecture developed based on member's common vision, and shared understanding of the future of service delivery, has defined the standards of separating management and operating concerns for organization objects, mapped them to business processes, implemented functional grouping void of any organization implementation specifics, and defined principles for decoupling and integration of architecture components. The evolution of the standards will be based on requirements that guides its continuity into the future systems.

This document represents the key artifacts that frames the Information Systems Architecture.

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1. Introduction

This document outlines the normative aspects of Open Digital Architecture Functional framework. It presents the Operations Map and the Technology Integration mapping along with descriptions and high level specifications for a digital platform and digital service provider. IT represents an agreed view of a majority of the industry key stakeholders strategic and operating plan.

The document targets CTIOs, Enterprise Architects, Platform Owners, Supply Chain Managers and Business Transformation officers. It can be used to define RFPs, refactor legacy BSS/OSS systems, as well as their management and operations. It defines standards for mapping business and information objects in a way that addresses concerns about vendor lock-in, product innovation, multi-vendor operating environment, etc.

As the Functional Architecture for the Open Digital Architecture, it enforces the principles established in GB998 Open Digital Architecture (ODA) Concepts & Principles and set out clarity for decoupling of functions, componentization of applications, and inclusion of systems of intelligence. Mapping to the Frameworx Business Process Framework (eTOM GB921) and Information Framework (SID GB922) are standards that capture business concerns of today, as well as new requirements that fulfill the changing dynamics of business, management and operations in a digital world.

1.1. Benefits of ODA

- Decouples Technical and Commercial evolution of the Business (It achieves this through a new Functional grouping which provides flexible composition and interaction)
- Provides direct and real-time cross-functional capability interaction between Systems of Engagement (SoE), Systems of Record (SoR) and Systems of Insights(SoI).
- Removes data replication between functions that work on the same data
- Allows playing different roles (Retailer, Biller, …)
- Exposes standardized APIs (TMF Open APIs, 3GPP SBIs, MEG APIs) at the border of each Functional block.
- Enforces event based microservice architecture
- Eliminates unnecessary bundling and functionality integration
- Ensure reusability

1.2. Scope of ODA Functional Architecture

- Normative Taxonomy of Business and Technology Functions
- Standard Functionality of a Digital Enterprise business Functions
- Standardized Core Business Processes for Functions (which can be extended by industry)
- Structured use of Frameworx.
- Technology neutral Features and Feature Grouping (using SID information Framework Definitions)

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That enables:

- Business to define requirements using an unambiguous set of Functions (Leverage same language as Frameworx for consistency in expressing requirements and reuse of function specifications).
- Creation of reusable Software component and specifications based on groupings of Functions
- Simplified Management and Operations through well-defined decoupling and integration principles

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2. The Functional Architecture of ODA

The ODA Function Blocks (below) define the key functional grouping that logically establish related family of information systems objects. Through functional decomposition of the key capabilities of a digital business/digital enterprise, the Functional Architecture establishes a Level 0 Functional grouping that enables achieves business and operational agility.

The Functional Architecture is modeled to represent digital enterprise business operation, using functions that correspond to the interactions that go beyond the traditional pipe-line business model. Each of the ODA Function Blocks is an enterprise level group of functions with a collection of coherent processes continuously performed to enable digital business.

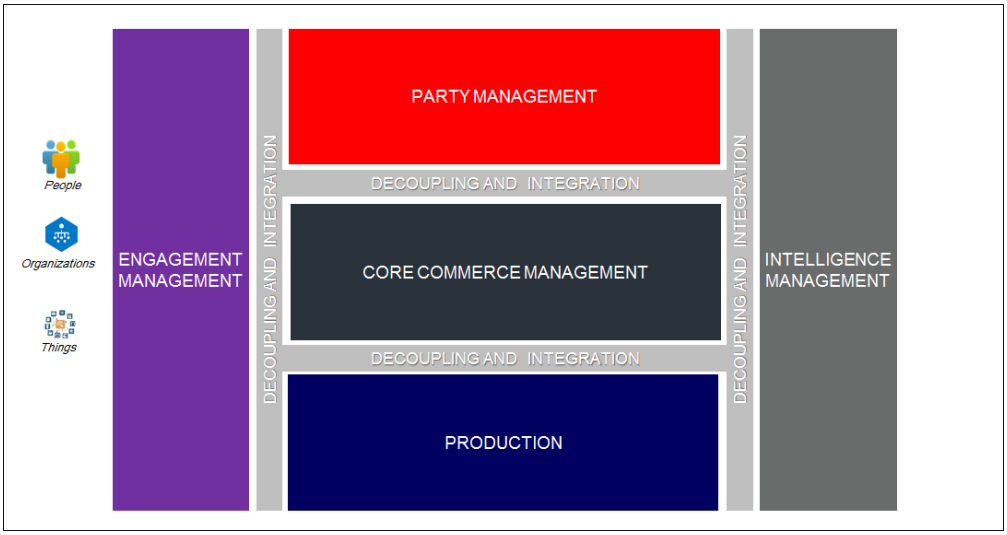

Figure 2.1. ***ODA Functional Architecture - High Level view (Level 0)***

As a Level-0 Functional Architecture, the Function Blocks can be used to quickly develop various information systems operating models of a specific enterprise based on its mission - while maintaining alignment integration to the overall vision of a digital enterprise.

Alternate View 1

An alternate view is provided in order to de-constrain the representation. The choice of view is not mandatory. Members can choose to change the view to fit any specific use, but the core standards must always remain consistent with the overarching principle - non-layered, the existence of a decoupling and integration function bordering all adjacent functions.

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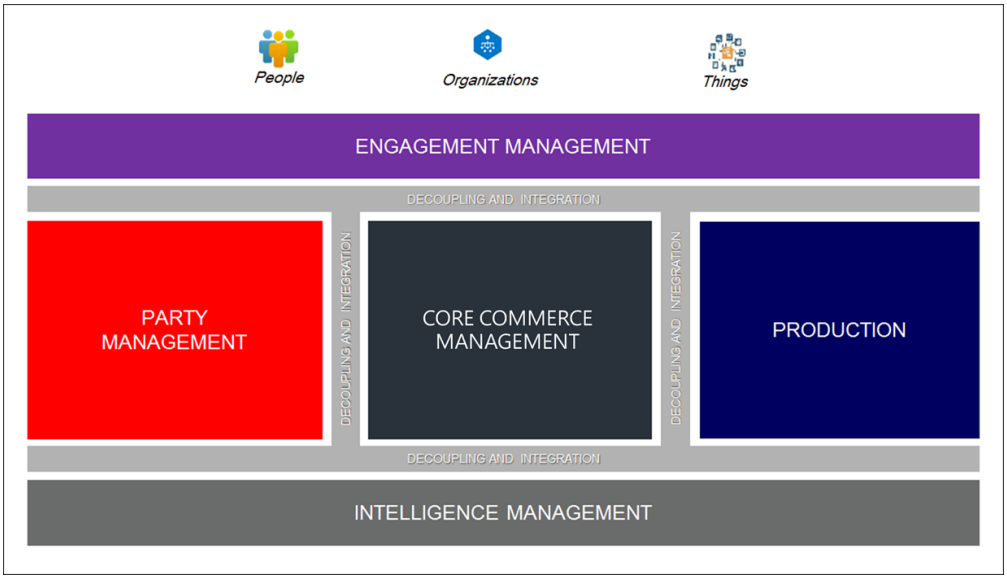

Figure 2.2. ***ODA Functional Architecture - alternate view (Level 0)***

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3. ODA Systems of Separation

The functional view of ODA can be summarized in a system view based on the architectural elements that deliver the system's functionality. The view in Figure 3-1 below groups the ODA function blocks into a system's structure-including the key functional elements (Party Management, Core Commerce Management, and Production), their responsibilities, which can be used to summarize their roles, interfaces they expose, and the interactions between them. Taken together, this shows how the responsibility of the systems and the functions required of them.

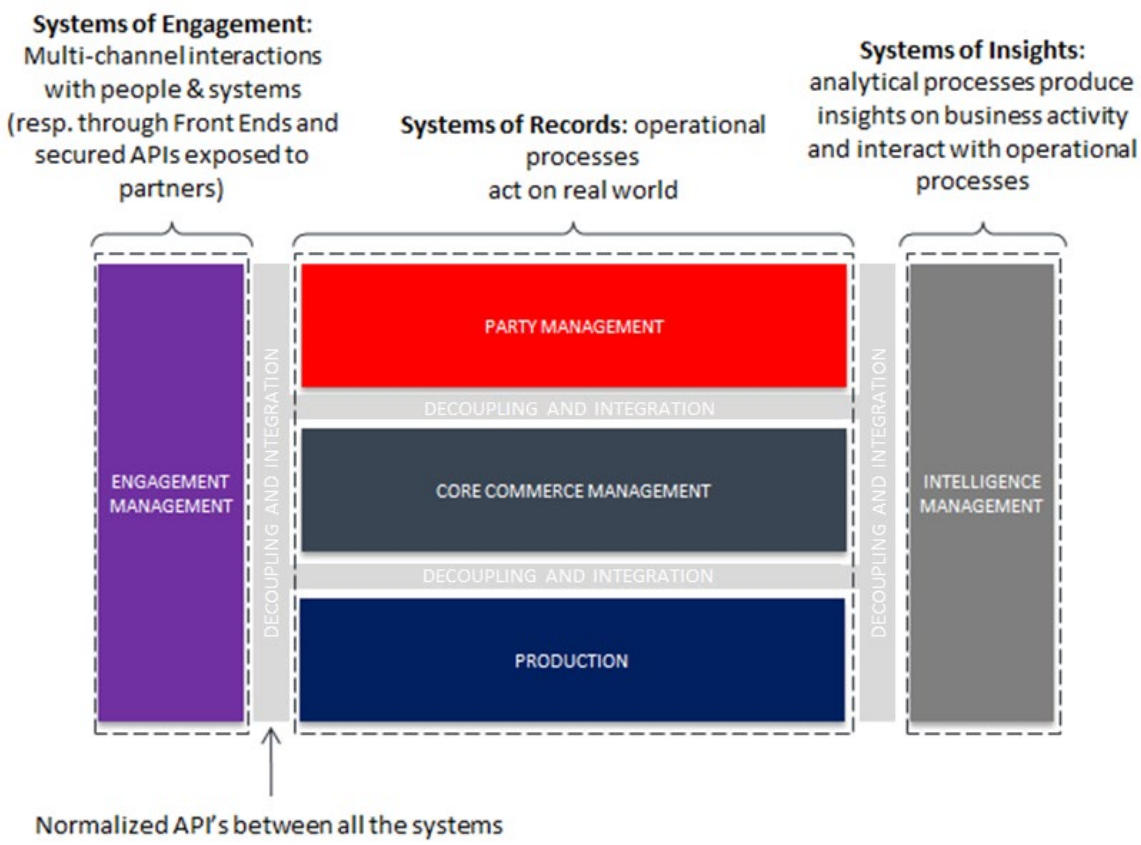

Figure 3.1. Grouping of systems based on the Functional Architecture

Rationale of systems separation

Each of above defined system types has its own lifecycle :

- **Engagement Management:**

  Responsible for user interactions that are powered by fast moving technologies, it is committed to evolve (e.g., in a quick & agile mode) in order to catch the trends on the fields of customer experience, ergonomics, social networking. These trends apply to all industries not only telecommunications, therefore, Service Providers have no control over them and must adapt quickly.

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- **Systems of Records**

  Responsible for operational processes, it consists of assets (processes, objects and data) that are core to the stability of an enterprise.

- **Systems of Insight**

  Responsible for knowledge-defined automation. It makes use of Big & Fast Data to enable cross-functional Intelligence functions and cognitive workflows.

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4. The Functional Blocks

4.1. Decoupling & Integration

4.1.1. Definition & Scope

The Decoupling & Integration (D&I) Block is about governing and managing separation of functional borders based on well-established "families" of closely interrelated services as well as the integration between them. D&I gathers non-business functions that are needed at design and runtime to enable the combinations of services that the business requires.

The Decoupling & Integration Block ensures that services provided by ODA can be combined in a flexible way, without restriction imposed by a process pattern derived from a given stack of blocks.

Note: Where in traditional architectures functional layers are supposed to communicate only with adjacent ones (e.g., BSS with OSS, OSS with Network but not BSS directly with Network). "Decoupling" within ODA emphasizes that ODA Function Blocks are not behaving like layers and tightly coupled together. This ensures that the functional architecture by itself doesn't preclude any combination of services.

4.1.2. Functions of Decoupling & Integration

Core functionalities covered by the "Decoupling and Integration" aspect of the functional architecture are outline in the table below.

Table 4-1. ***Decoupling & Integration key functionalities***

#	Functionality	Description / Purpose
DI.001	Normalized APIs	Catalog of API's - Management of API catalog and documentations, including links to catalogs of partners
DI.002	Message Routing	API Routing - Route API calls to endpoint, API Mediation
	...	

4.2. Engagement Management

4.2.1. Definition & Scope

The Engagement Management (EGM) Block represents the interaction point between the enterprise and its ecosystem, which can be through different channels for different experiences. EGM provides the capabilities to assure a digital, secure and accurate interaction of all these stakeholders, providing ***digital and omnichannel capabilities*** through the configuration of the different journeys reusing the processes and functionalities provided by the platform.

Its scope includes interactions between users and processes handled by other functions such as from the Intelligence Management, Party Management, Core Commerce Management or Production Blocks. “Users” from the perspective of EGM, may be machines and things (IoT

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context), humans (prospects, customers, users, employees, of people working on behalf of partners and suppliers - 3rd parties) that collaborate and integrate their own solutions to the ecosystem, providers of different services (retailers, salespeople, technical providers).

4.2.2. Engagement Management is characterized as:

- Handling relationship between external and internal actors (including 3rd party systems, connected objects, …) and other ODA blocks.
- Managing the presentation layer according to the channel, the device to allow multi and cross-channel user journeys over multiple types of devices.
- Tailoring interactions using contextual information from back ends (360°view, process states).
- Relying on Processes, Functions and Data stored in the other ODA blocks (this is what enables reuse, cross channel and multi device).
- "Powering" customer / user journeys (what users see of the ongoing back end processes).
- Holding no business process (eTOM), no functions (functional Framework) and no operational data (SID)
- Holding technical functions – and data related to these functions and to their execution (context)
- Interacting with other ODA blocks only through Event or Process APIs (new APIs - definition ongoing), or read data through current standard TMF Open API

4.2.3. Engagement Management Block functionalities

This is the list of standardized Engagement Management functionalities. As new standards emerge, they will be updated into the table below.

Table 4-2. ***Functionalities provided by the Engagement Management Block include:***

#	Functionality	Description / Purpose
EM.001	Front-ends (UIs)	Portals and application interfaces that are rendered for interactions.
EM.002	Journey Management	Configure and adapt interactions to different types of Parties, Channels etc.
EM.003	Access to content	Access in read only to data contained in different repositories, according to Party rights. It can include access through Front-Ends or via APIs.
EM.004	Content Aggregation	Collects information from multiple internal and / or external sources, including digital body language, and transcodes them for a single presentation format, storing document references.
EM.005	Content Organization	Classifies or categorizes data according to predefined ranking plans

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#	Functionality	Description / Purpose
EM.006	Personalization of content	Collects user's preferences and personalization criteria. Adapt contents and renders them based on context, user, interfaces, and according to user's preferences and personalization criteria.
EM.007	Filtering of content	Provides select capabilities for content based on engagement context and criteria.
EM.008	Information Services	Selective notification of information or alerts based on events (including social listening) towards the appropriate Actors, channel and device.
EM.009	User Interface Orchestration	Defines, formats and structures the sequence of user interfaces (UIs) needed by a process.
EM.010	Integration	Integrating internal systems/software/devices with external systems/software/devices - this includes social network platforms etc.
EM.011	API HUB	Exposes standardized API to Partners and external systems. Manages security and checks Access right policies.
EM.012	Authentication and Authorization*	Correctly authenticates and authorizes a Party, including role management.
EM.013	User-interaction lifecycle management*	Manage Party interactions with the different services (provide consent, configure services, etc.) as well as their relationship with other users (create groups and share services)

**These functionalities are still under review for confirmation.*

4.2.4. Open API's Mapping

#	Interface Name	Description/Purpose	Specification / Reference
1	Process flow	Engagement management doesn't have the right to Create, Delete, Update data. So, it cannot use the current TMF Open APIs, which are CRUD operations on data. The Process flow API is defined to enable Systems of Records and Intelligence to invoke kinematics in the Engagement Management function.	Open API TMF701
2	Event Management		Open API TMF688

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4.3. Party Management

4.3.1. Definition & Scope

The Party Management (PM) Block handles all interactions and data associated with entities or actors that are involved or likely to get involved with the enterprise. Party roles and behaviors are not dependent on industry specifics. This block includes support for identification processes which are required to validate parties and management of Party relationships.

Party Management can be characterized as follows:

- Party Management deals with "WHO" & "WHY"
- It handles Persons and Organizations (called Party) involved in business processes. These Parties can be Customers, Business Partners or Employees.
- In charge of Party oriented processes, functions and repositories such as:
    - Party, Party Information and Privacy Management, Party Roles and Rights Management
    - Party Interaction Management
    - Billing Account Management and Bill Production
    - Party Financial Management activities : payment, bill inquiry, dunning, …
    - Market and Party Strategy activities
    - etc.
- Party Management is not Telco specific, as any type of digital industry has now this kind of needs.

4.3.2. Business Processes in the Party Management Block

Party Management mapping is based on a tactical operating model with no categoric business architecture underpinning it. This mapping is provided for informative purposes only, where different Business Architecture shall have an influence on eventual mapping.

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Business Process entities and their descriptions (Source GB921 R20.5)

Table 4-3. ***Party Management eTOM Process Descriptions***

Process Identifier	eTOM Process	Category	Brief Description
1.1.1	Market Strategy & Policy	(2) eTOM Process Type	Enable the development of a strategic view of an enterprise's existing and desired market-place
1.1.2	Sales Strategy & Planning	(2) eTOM Process Type	Develop an appropriate sales strategy to complement the market strategy.
1.1.3	Sales Forecasting	(2) eTOM Process Type	Sales Forecasting involves gathering current and historic order information, analyzing sales trends and patterns generating sales forecast and analyzing historical and planned promotions and events.
1.1.5	Sales Development	(2) eTOM Process Type	Develop the Sales support and response for new and existing products, as well as existing and potential customers.
1.1.8	Sales Channel Management	(2) eTOM Process Type	This process element represents part of the overall enterprise, modeled in business process terms, and can be applied (i.e., "instantiated") with other similar process elements for application within a specific organization or domain.
1.1.9	Selling	(2) eTOM Process Type	Responsible for managing prospective customers, for qualifying and educating customers, and matching customer expectations Managing prospective parties with whom an enterprise may do business, such as potential existing or new customers and partners, for qualifying and educating them, and ensuring their expectations are met.
1.1.11	Contact/Lead/Prospect Management	(2) eTOM Process Type	Develop the appropriate relationships with contacts, leads, and prospects with the intent to convert them to consumers, such as customers, or

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Process Identifier	eTOM Process	Category	Brief Description
			providers, such as partners, of an enterprise's offerings.
1.1.14	Marketing Communications and Advertising	(2) eTOM Process Type	Develop and manage communications to the market, prospective and existing customers. Communications involve both the message and the media. Develop a message and manage its delivery. Also develop and manage interfaces with press/news and manage an editorial calendar to plan placements.
1.1.16	Brand Management	(2) eTOM Process Type	Create a name, symbol or design that identifies and differentiates an enterprise's product offering and/or offerings from its others product offering(s) and similar offerings available from other enterprises, including competitors.
1.1.17	Market Research	(2) eTOM Process Type	Gathers, analyze and interpret information about a market, about product offerings that will be made to the market, and about the past, present and potential customers for the offering; Research the characteristics, such as spending habits and a target market's requirements, the industry as a whole, and competitors.
1.3.2	Customer Experience Management	(2) eTOM Process Type	Perform customer lifecycle mapping, assess customer experience maturity, as well as analyze customer experience metrics.
1.3.4	Customer Management	(2) eTOM Process Type	Manage the relationship of the Customer and the enterprise.
1.3.5	Customer Interaction Management	(2) eTOM Process Type	Manage interactions between the customer and the enterprise. Interactions can be triggered by the customer or by the enterprise

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Process Identifier	eTOM Process	Category	Brief Description
1.3.6	Customer Information Management	(2) eTOM Process Type	Manage customer information after customer contracts or associated service orders have been finalized and during the order completion phase. Ensure that any customer information required by other CRM processes is updated as part of the customer order completion.
1.3.9	Customer Bill Invoice Management	(2) eTOM Process Type	Ensure the bill invoice is created, physically and/or electronically produced and distributed to customers, and that the appropriate taxes, discounts, adjustments, rebates and credits for the products and services delivered to customers have been applied.
1.3.10	Customer Bill Payments & Receivables Management	(2) eTOM Process Type	Ensure that enterprise revenue is collected through pre-established collection channels and put in place procedures to recover past due payments.
1.3.11	Customer Bill Inquiry Handling	(2) eTOM Process Type	Ensure the timely and effective fulfillment of all customer bill inquiries and complaints.
1.3.15	Customer Experience Management Strategy and Planning	(2) eTOM Process Type	This set of processes will enable the development of a strategy and plan for how customer experience management would be incorporated into the who organization.
1.3.16	Customer Inventory Management	(2) eTOM Process Type	Establish, manage and administer the enterprise's customer inventory, as embodied in the Customer Inventory Database, and monitor and report on the usage and access to the customer inventory, and the quality of the data maintained in it.
1.6.1	Party Strategy & Planning	(2) eTOM Process Type	Develop the strategies and policies of the enterprise for engagement with other parties

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Process Identifier	eTOM Process	Category	Brief Description
1.6.2	Party Tender Management	(2) eTOM Process Type	Party Tender Management processes manage the activities associated with developing tender documents, running tender processes, gaining enterprise agreement to tender decisions.
1.6.3	Party Relationship Development & Retirement	(2) eTOM Process Type	Manage the lifecycles of parties with whom the enterprise has a relationship. Relationship with new parties may be required to broaden the services an enterprise offers, to improve performance, for outsourcing and out-tasking requirements, and so forth.
1.6.5	Party Agreement Management	(2) eTOM Process Type	Manage the evaluation of agreements with parties, including customers. Initiate and complete business agreements when one or more other parties are involved.
1.6.7	Party Privacy Management	(2) eTOM Process Type	Manage the Privacy of Data Subject Party Information in accordance with privacy regulations and the Data Subject's explicit wishes. Create and define the Privacy Profile Type, manage the definition of the Privacy Profile with the Data Subject Party as well as the evolution of this profile if required, and ensure that the Data Controller Party can demonstrate compliance by itself and any other Party.
1.6.9	Party Interaction Management	(2) eTOM Process Type	Manage interactions between parties and the enterprise. Interactions can be triggered by the enterprise (as a result of a query or complaint) or by a party (for example sending bills or other notifications.)
1.6.13	Party Training and Education	(2) eTOM Process Type	Assess an enterprise's training needs, design training plans, develop training, conduct and evaluate training and results.

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Process Identifier	eTOM Process	Category	Brief Description
1.6.14	Party Special Event Management	(2) eTOM Process Type	Plans, prepare, and produce a special event which targets one or more types of parties. It includes the assessment, definition, acquisition, allocation, direction, control, and analysis of time, finances, people, products, services, and other resources to achieve an event's objectives.
1.6.15	Business Partner Bill/Invoice Management	(2) eTOM Process Type	Manage the party bill/invoice process, control bills/invoices, manage the lifecycle of bills/invoices, and perform bill/invoice trend analysis. A bill is a notice for payment which is supposed to be preceded by an invoice in most cases. An invoice typically does not summarize to total outstanding payment or payments due. A bill typically contains the payment due date and balance.
1.6.16	Business Partner Bill Payments & Receivables Management	(2) eTOM Process Type	Party Bill Payments & Receivables Management is responsible for management of methods used by parties to make payments, administers payment plans, handles payments, and collects debt.
1.6.20	Business Partner Bill Inquiry Handling	(2) eTOM Process Type	Ensure the timely and effective resolution of all party bill inquiries and complaints initiated by an enterprise or initiated by another party, such as partner.
1.6.21	Party Inventory Management	(2) eTOM Process Type	Manage the administration of the enterprise's party inventory.
1.X.Y	Party Identification & Authentication	(2) eTOM Process Type	*to be discussed for next versions*

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4.3.3. Standard Business Information Entities

Below is a mapping of Frameworx SID ABEs to the Party Management functional block

Table 4-4. ***SID ABEs and their description (Source GB922 R20.5)***

SID Domain	#	ABE	ABE Definition
Market / Sales	01	Market & Sales Strategy and Plan	Supports the business plans and strategies on how to address the market with appropriate products and channels.
	02	Market Segment	Supports market segments, market statistics, and forecasts
	03	Competitor	Identifies other providers who compete in the same market segments, accumulates intelligence about the competitors, including products (price, Key Performance Indicators and so forth).
	04	Contact / Lead / Prospect	Provides the ability to track sales leads through their life cycle up until the time the leads become customers, including lead and contact information, sales prospects, proposals made to potential customers, and the amount of potential revenue the leads represent in the form of a sales pipeline.
	05	Sales Channel	Keeps track of distribution channels and sales activities, sales quotas, sales, contests, commission/bonus plans, commissions/bonuses, and maintain groups of individuals that make up the sales force.

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SID Domain	#	ABE	ABE Definition
	06	Market Sales Forecast	Maintains market and sales forecasts, new service requirements, customer needs, and customer education.
	07	Market Sales Party Roles	Market Sales Party Roles ABE contains all PartyRoles related to the Market Sales Domain such as MarketingManager, SalesAgent...
Customer	01	Customer Party Roles	The Customer Party Roles ABE is the focus on all the roles related to Customer.
	02	Customer Interaction	Represents communications with customers, and the translation of customer requests and inquiries into appropriate “events” such as the creation of a customer order, the creation of a customer bill inquiry, or the creation of a customer problem.
	03	Customer Bill	Handles real-time and non-real-time Call Detail Records (CDRs) and other sources of data that result in invoice items. The Customer Bill ABE also represents the format of a bill, schedule the production of bills, customer invoicing profiles, all the financial calculations necessary to determine the total of the bill (except for rating and rating discounts), and credits and adjustments to bills.
	04	Customer Bill Collection	Handles credit violations, actions for overdue debts, and facility billing audits.
	05	Customer Bill Inquiry	Represents invoice inquiries associated with invoices sent to customers and handles disputes and adjustments on individual charges, invoices, and accounts.
Business Partner	01	Party Strategy & Plan	The strategies and the planning of the business relation with the other parties with input from other ABEs, such as MarketSales, Party Performance and Competitor Analysis.
	02	Party	A Party represents an individual or an organization, or an organization unit that are of interest, involved or that provide value directly or indirectly from an enterprise perspective (Enterprise is to be understood here as provider, service provider or operator). Hence, the Party plays one or more roles with the enterprise or with another Party playing a role with the enterprise (indirectly). This is

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SID Domain	#	ABE	ABE Definition
			introduced to specify that the only party an enterprise would be interested in and will consider in its systems is a party playing a role (directly or indirectly). Roles would be represented by the Party Role concept. Additional roles will be defined to cover the needs of the new digital ecosystem." Examples of roles played by Parties include those: • Of interest, such as competitors, sales prospects • Involved, such as customers, users, employees • That provides value directly or indirectly, such as service providers, operators, cloud brokers, infrastructure providers, application developers.
	03	Business Partner Account	Additional Party Entities ABE represents entities that are similar to those in the Customer ABE, such as PartyAccount and Party Account Contact.
	04	Party Bill	*(preliminary)*
	05	Party Bill Collection	The Party Bill Collection ABE handles credit violations, actions for overdue debts (Dunning), and all what is related to PartyPayment.
	06	Business Partner Party Roles	Manages Partner roles and related information.
	07	Party Interaction	Represents communications with parties, and the translation of requests and inquiries into appropriate "events" such as the creation of an order, the creation of a bill inquiry, or the creation of a problem.
	08	Party Privacy	Contains all entities used by the Party Privacy Management process for specifying the information concerned by Privacy rules, the Privacy rules themselves, and the choices made by Parties for their own Privacy.
Common	01	Communication Interaction	The Communication Interaction ABE contains all entities needed for specifying a CommunicationInteraction describing an exchange of information during a communication between two or more human(s) and/or machine(s) during a period of time.

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SID Domain	#	ABE	ABE Definition
	02	Users and Roles	Represents types of users and roles and their involvement in the usage of products, services and resources.
Enterprise	01	Enterprise Party Roles	Enterprise Party Roles ABE contains all PartyRoles related to the Enterprise such as Employee, Employer, Regulator...
Product	01	Product Party Roles	Product Party Roles ABE contains all PartyRoles related to the Product Domain such as ProductManager
Service	01	Service Party Roles	Service Party Roles ABE contains all PartyRoles related to the Service Domain such as ServiceManager.
Resource	01	Resource Party Roles	Resource Party Roles ABE contains all PartyRoles related to the Resource Domain such as Technician, ResourceManager.

4.3.4. Standard Interfaces

List of Open APIs proposed to be provided by Party Management functional bloc.

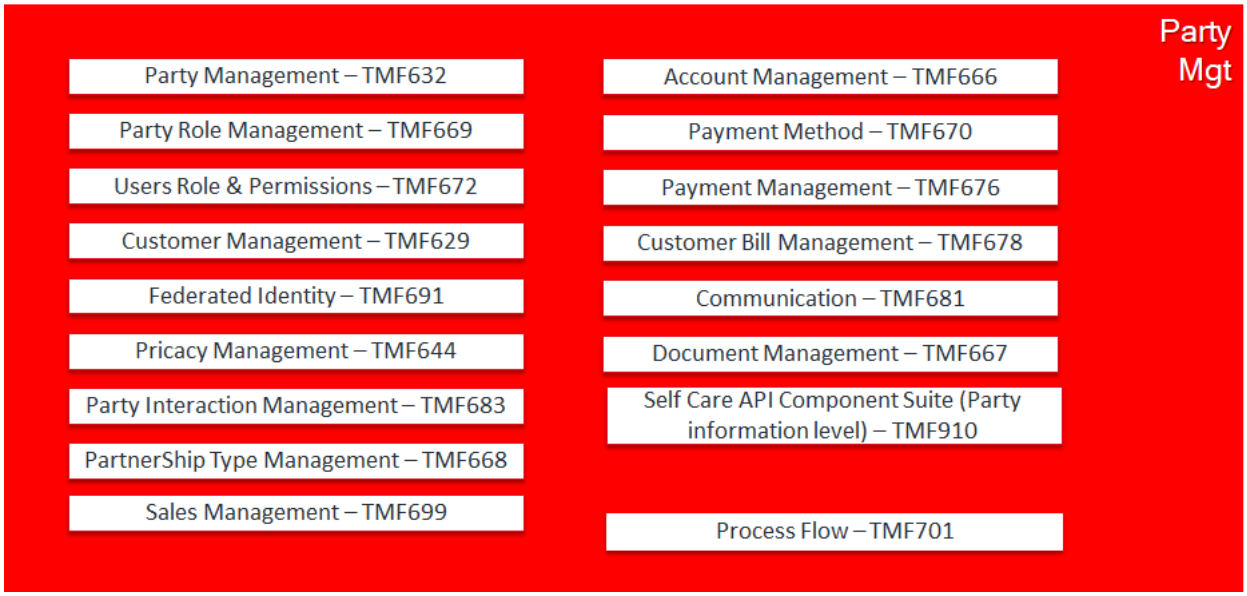

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Table 4-5. ***Proposed Standard Interfaces for Party Management block***

#	Interface Name	Purpose/Description	Specification / Reference
1	Party Management API		Open API TMF632
2	Customer Management API		Open API TMF629
3	Privacy Management API		Open API TMF644
4	Party Role Management API		Open API TMF669
5	Account Management API		Open API TMF666
6	Document Management API		Open API TMF667
7	Partnership Type Management API		Open API TMF668
8	Payment Method API		Open API TMF670
9	Users Role & Permissions API		Open API TMF672
10	Payment Management API		Open API TMF676
11	Customer Bill Management API		Open API TMF678
12	Communication API		Open API TMF681
13	Party Interaction Management API		Open API TMF683
14	Federated Identity API		Open API TMF691
15	Sales Management API		Open API TMF699
16	Process Flow API		Open API TMF701
17	Self-Care API Component Suite (Party information level)		Open API TMF910

4.4. Core Commerce Management

4.4.1. Definition & Scope

The Core Commerce Management (CCM) Block represents the part of the enterprise that is concerned with enabling profitable exchange of goods and services. It includes all those activities which directly or indirectly facilitate that exchange such as marketing & sales, sourcing and procurement, strategic (buying & selling) portfolio plans (commercial rules), life-cycle product management, offer management, new business development - for all types of business engagements - B2B, B2C, C2C, B2B2X etc. It is responsible for formalizing business models, revenue generation and business assurance processes.

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Core Commerce Management can be characterized as follows:

- Core Commerce Management deals with "WHAT"
- It handles the catalogue of Product Offerings and Products marketed by the telco operator for all types of business (B2C, B2B, B2B2X, …)
- Core Commerce Management is in charge of Product Offerings and Products Catalogue oriented processes, functions and repositories such as :
    - Product Offerings and Products Catalogue Management
    - Order Handling, from Order Capture & Configuration to the global orchestration of the Customer Order Delivery
    - Rating of any types of charges and Bill items Calculation
    - Business Partners Revenue Sharing & Settlement
    - Product Assurance
    - Product Strategy activities
    - Etc.
- Core Commerce Management is independent from technical concerns - managed by Production – to be able to decouple commercial and technical evolution

Core Commerce Management Block functionalities

4.4.2. Business Processes in the Core Commerce Management Block

The Core Commerce Management Block mapping is based on a tactical operating model with no categoric business architecture underpinning it. This mapping is provided for informative purposes only, where different Business Architecture shall have an influence on eventual mapping. This mapping is based on eTOM level 2 business processes.

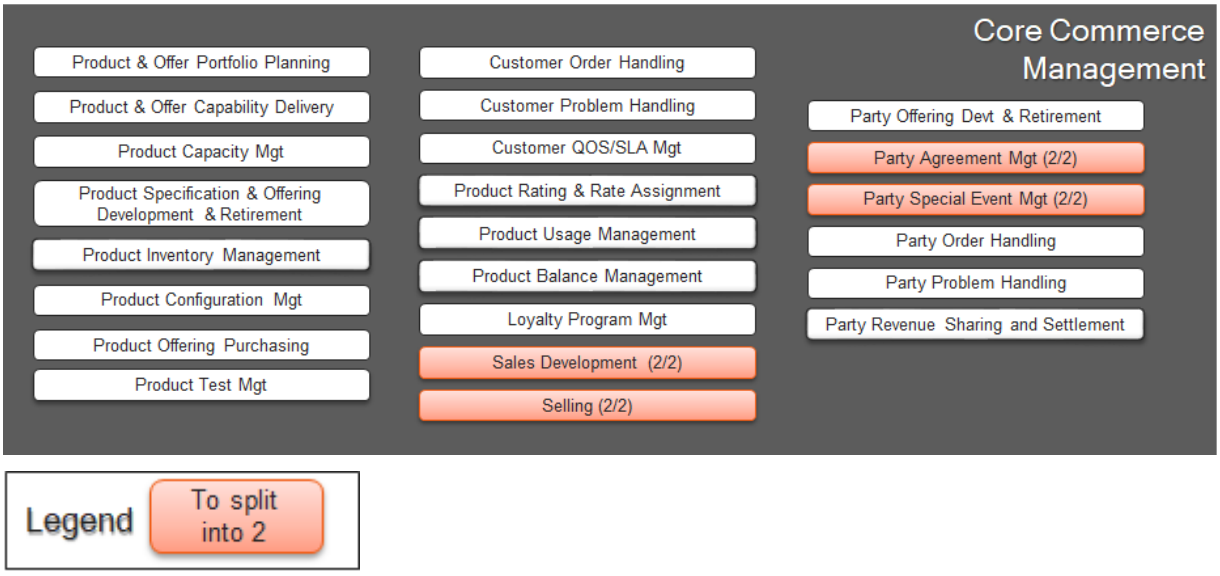

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Table 4-6. ***Core Commerce Management eTOM Process Descriptions (Source reference GB921 R20.5)***

Process Identifier	eTOM Process	Category	Brief Description
1.1.5	Sales Development	(2) eTOM Process Type	Develop the Sales support and response for new and existing products, as well as existing and potential customers.
1.1.9	Selling	(2) eTOM Process Type	Responsible for managing prospective customers, for qualifying and educating customers, and matching customer expectations Managing prospective parties with whom an enterprise may do business, such as potential existing or new customers and partners, for qualifying and educating them, and ensuring their expectations are met.
1.1.19	Loyalty Program Management	(2) eTOM Process Type	Define all aspects of a loyalty program, such as requirements and objectives of a loyalty program, determine the benefits to participants. Develop a program, prototype it, test it, roll out/launch it, amend and evaluate it, and terminate it when it is no longer viable for an enterprise. Manage all operational aspects of running a loyalty program. Enable parties to become members of a program, earn currency and rewards, and redeem currency. Manage a loyalty program account, leave a program, and provide operational reports.
1.2.1	Product & Offer Portfolio Planning	(2) eTOM Process Type	Develop strategies for products at the portfolio level.
1.2.2	Product & Offer Capability Delivery	(2) eTOM Process Type	Manage the delivery and build of new or changed Product & Offer and delivery capabilities within an enterprise.
1.2.5	Product Configuration Management	(2) eTOM Process Type	Product Configuration involves specifying how a Product operates or functions in terms of a product's configurable properties and features, collectively called characteristics, and related products, services, and resources that are used in its configuration.

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Process Identifier	eTOM Process	Category	Brief Description
			Specify how a product operates or functions in terms of a product's configurable parameters, collectively called characteristics, and related configurations of products, services, and resources that are part of its overall configuration. The configuration for a product, which is also managed by this process, is governed by a selected product configuration specification.
1.2.7	Product Specification & Offering Development & Retirement	(2) eTOM Process Type	Develop and deliver new product specifications as well as enhancements and new features, ready for use by other processes, including Product Offering Development & Retirement. Develop and deliver new product offerings, their pricing, as well as catalogs that contain both.
1.2.8	Product Capacity Management	(2) eTOM Process Type	Specify available product capacity and capture product capacity demand for defined time periods. The consumption of available capacity by demand is captured for a given time period.
1.2.9	Product Offering Purchasing	(2) eTOM Process Type	Make an inbound/outbound purchase of one or more product offerings, change an offering being purchased, review an entire purchase, and other processes that manage the lifecycle of a purchase of one or more product offerings.
1.2.11	Product Inventory Management	(2) eTOM Process Type	Establish, manage and administer the enterprise's product inventory, as embodied in the Product Inventory repository, and monitor and report on the usage and access to the product inventory, and the quality of the information maintained in it.
1.2.15	Product Test Management	(2) eTOM Process Type	Product Test Management provides the manages the end-to-end execution of a test or test scenario for products not specific to a customer.

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Process Identifier	eTOM Process	Category	Brief Description
1.2.16	Product Usage Management	(2) eTOM Process Type	Encompasses the functions required to guide, distribute, mediate, summarize, accumulate, and analyze Product Usage records.
1.2.17	Product Rating & Rate Assignment	(2) eTOM Process Type	Rate a value (monetary or other) to a Product Usage or a set of Product Usages and assign the result to a Product and a Billing Account.
1.2.18	Product Balance Management	(2) eTOM Process Type	Management of Product balances.
1.3.3	Customer Order Handling	(2) eTOM Process Type	Responsible for accepting and issuing orders.
1.3.7	Customer Problem Handling	(2) eTOM Process Type	Responsible for receiving trouble reports from customers, resolving them to the customer's satisfaction and providing meaningful status on repair and/or restoration activity to the customer.
1.3.8	Customer QoS/SLA Management	(2) eTOM Process Type	Monitoring, managing and reporting of delivered vs. contractual Quality of Service (QoS), as defined in the enterprise's service descriptions, customer contracts or product catalogue.
1.6.4	Party Offering Development & Retirement	(2) eTOM Process Type	Manage on-boarding and off-boarding another party's product specifications and product offerings that a required to facilitate the business model of the enterprise.
1.6.5	Party Agreement Management	(2) eTOM Process Type	Manage the evaluation of agreements with parties, including customers. Initiate and complete business agreements when one or more other parties are involved.
1.6.8	Party Order Handling	(2) eTOM Process Type	Track, monitor and report on an order to another party to ensure that the interactions are in accordance with the agreed commercial agreements with the other party.
1.6.10	Party Problem Handling	(2) eTOM Process Type	Ensure the timely and effective resolution of all party problems initiated by an enterprise or initiated by another party, such as a partner.

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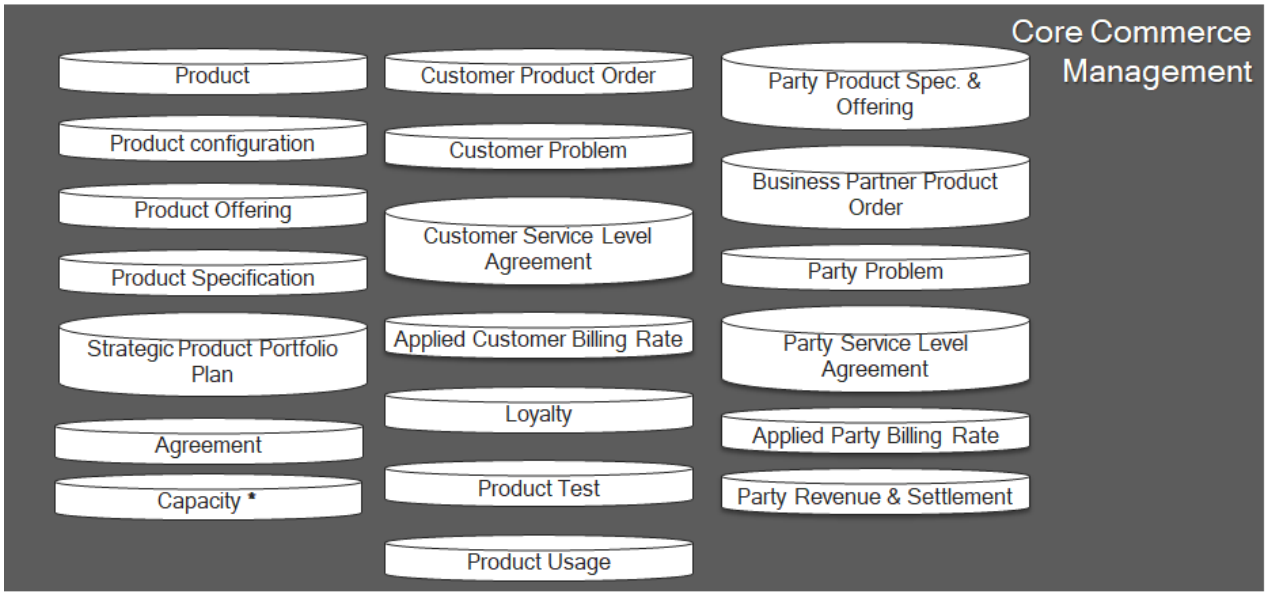
Process Identifier	eTOM Process	Category	Brief Description
1.6.14	Party Special Event Management	(2) eTOM Process Type	Plans, prepare, and produce a special event which targets one or more types of parties. It includes the assessment, definition, acquisition, allocation, direction, control, and analysis of time, finances, people, products, services, and other resources to achieve an event's objectives.
1.6.18	Party Revenue Sharing and Settlements	(2) eTOM Process Type	Develops party revenue sharing models, prepare party revenue sharing agreements, determine party revenue shares, and reconcile party revenue shares.

4.4.3. Standard Business Information Objects

Below is a mapping of Frameworx SID ABEs to the Core Commerce Management functional block



* At productlevel - Shared with Production for Service et Resource levels – further study required

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Table 4-7. ***SID ABEs and their description (Source GB922 R20.5)***

SID Domain	#	SID ABE	ABE Definition
Customer	01	Customer Product Order ABE	Handles single customer orders and the various types thereof, such as regulated and non-regulated orders.
	02	Customer Problem ABE	Focuses on technical assistance and problem handling for customers.
	03	Customer SLA ABE	Is a special case of the Service Level Agreement ABE where an involved party in the agreement is a Customer. See the Agreement ABE in the Engaged Party Domain for details.
	04	Applied Customer Billing Rate ABE	Deals with the correlation of related usage for subsequent rating, rates applied to the usage (both regulated and non-regulated), discounts to usage, and any taxes due on the rated usage.
Product	01	Product ABE	Represents an instance of a product offering subscribed to by a party, such as a customer, the place where the product is in use, as well as configuration characteristics, such as assigned telephone numbers and internet addresses. The Product ABE also tracks the services and/or resources through which the product is realized.
	02	Product Specification ABE	Contains all what represent a product specification as perceived by the business user and specifies what the marketing operator wants to sell at a functional level (i.e., what are the capacities, which usages are available…).
	03	Strategic Product Portfolio Plan ABE	Is concerned with the plans of the product portfolio, which product offerings to make available to each market segment and the plans to develop and deploy product offerings, as well as retirement of products
	04	Product Offering ABE	Describes tangible and intangible goods made available for a certain price to the market in the form of product catalogs. This ABE is also responsible for targeting market segments based on the appropriate market strategy.
	05	Product Usage ABE	Represents usage of products associated with Customers used for charging that may appear on a Customer Bill.

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SID Domain	#	SID ABE	ABE Definition
	06	Product Configuration ABE	The definition of how a Product operates or functions in terms of CharacteristicSpecification(s) and related ResourceSpec(s), ProductSpec(s), ServiceSpec(s) as well as a representation of how a Product operates or functions in terms of characteristics and related Resource(s), Product(s), Service(s).
	07	Loyalty ABE	A loyalty program is one of the tools used by the loyalty process to retain customers. The Loyalty ABE contains all entities useful to specify and instantiate loyalty programs. A LoyaltyProgramProdSpec defines the LoyaltyRules that have to be checked in order to identify the actions to apply. Depending on the type of LoyaltyRules a LoyaltyAccount might be needed to collect gains or not. A LoyaltyProgramProduct is a type of ProductComponent and described by a LoyaltyProgramProdSpec.
	08	Product Test ABE	A product test is a function performed on a product that results in measures being produced that reflect the functioning of the entity under test.
Business Partner	01	Party Problem ABE	Focuses on technical assistance and problem handling reported to and by other parties.
	02	Business Partner Product Order ABE	Contains all entities required to specify a communication used to procure or update one or many Products in the context of a ProductOffering.
	03	Party Product Specification & Offering ABE	Represents the involvement parties playing roles have with ProductSpecifications and Product Offerings, such as ordered from and billed by.
	04	Party Revenue Settlement	Party Revenue ABE is composed of revenue sharing and settlements.
	05	Party Service Level Agreement	Party Service Level Agreement ABE represents an SLA with one or more other parties. See the Agreement ABE for additional details.
	06	Applied Party Billing Rate	*(Preliminary - Ongoing study)*
Common	01	Agreement ABE	Represents a contract or arrangement, either written or verbal and sometimes enforceable by law, such as a service level agreement or a customer price agreement. An agreement involves a number of other

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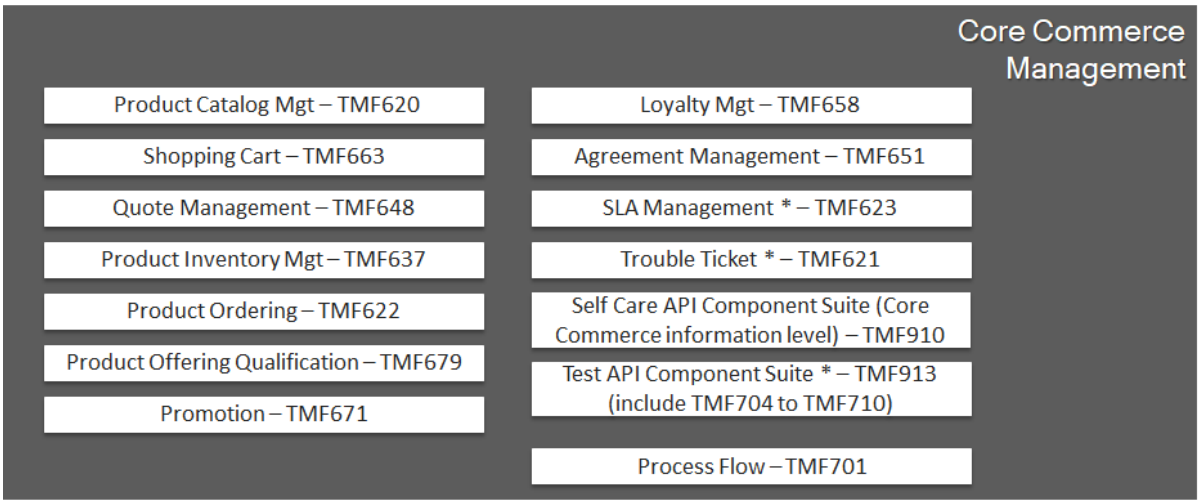
SID Domain	#	SID ABE	ABE Definition
			business entities, such as products, services, and resources and/or their specifications.
	02	Capacity ABE	A generalized ABE and specializations that provide capacity and capacity demand for products, services and resources

4.4.4. Standard Interfaces



* Shared with Production for Service & Resource levels – further study required

Table 4-8. ***Standard Interfaces in the Core Commerce block***

#	Interface Name	Purpose/Description	Specification / Reference
1	Product Catalog Management API		Open API TMF620
2	Trouble Ticket API		Open API TMF621
3	Product Ordering API		Open API TMF622
4	SLA Management API		Open API TMF623
5	Product Inventory Management API		Open API TMF637
6	Quote Management API		Open API TMF648
7	Agreement Management API		Open API TMF651
8	Loyalty Management API		Open API TMF658
9	Shopping Cart API		Open API TMF663
10	Promotion API		Open API TMF671

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#	Interface Name	Purpose/Description	Specification / Reference
11	Product Offering Qualification API		Open API TMF679
12	Process Flow API		Open API TMF701
13	Self-Care API Component Suite (Core Commerce information level)		Open API TMF910
14	Test API Component Suite (Product level)		Open API TMF913 (includes TMF704 to TMF710)

4.5. Production

4.5.1. Definition & Scope

The Production Block is responsible for the delivery and lifecycle management of Customer Facing Services (CFS) and Resource Facing Services (RFS) regardless of the technology type (e.g., physical, virtual, connectivity, end point, etc.) or the operational domain or factory where it originates, including third parties.

The Production Block exposes its service capabilities in a way that is commonly known as Network as a Service (NaaS) for consumption by other ODA function blocks. For example, the Core Commerce Management Block uses them to create products and offers, while the Engagement Management Block uses them to allow customers to recharge their balance or order a new video on demand. The Production block may also expose devices (e.g., smartphone) and application capabilities (e.g., email).

The scope of the Production block includes the exposure of its service capability definitions and the dynamic run-time decision of how the service(s) supporting Products and Customer will be created, delivered, used, maintained, assured, and repaired. It provides end-to-end management of operational functions for services and resources to support the commercial success of the ecosystem - including activities that span multiple enterprises such as the management of installation, deployment, and operations of technologies, performance and quality.

The Production Block does not need to know the details of a product nor the customer, contract owner, or payer requesting its exposed service capabilities. It assumes responsibility for the delivery of the CFS/RFS, as well as for operating the usage functions up to a committed quality of service - including applying policy control and balance management and supplying the Core Commerce Management Block with usage data records to perform rating on offers.

Production can be characterized as follows:

- Production deals with "HOW "
- Production is in charge of Services and Resources oriented processes, functions and repositories such as:
    - Definition and lifecycle of the telco capabilities that can be exposed as Services (CFS) to define Products and Offers to commercialize (in Core Commerce Management)

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- Definition and lifecycle of the technical solutions available for these services (RFS), based on all type of technology (physical, virtual, …) or all network, service platforms, IT systems, … capabilities, directly managed by the telco operator or available by means of a Business Partner
- Infrastructure Deployment and Operations management
- Delivery of services and resources
- Usage of services, including usage control and real-time balance management
- Problem and Performance Management
- Service & Resource Strategy activities
- etc.

Business Processes in the Production Function

The Production Block's business processes are based on a tactical operating model with no business architecture lock-in.

The business processes are derived from eTOM level 2 business process elements. The processes descriptions are derived from the following eTOM Groupings: Service Development and Management, Service Management and Operations, Resource Development and Management and Resource Management and Operations

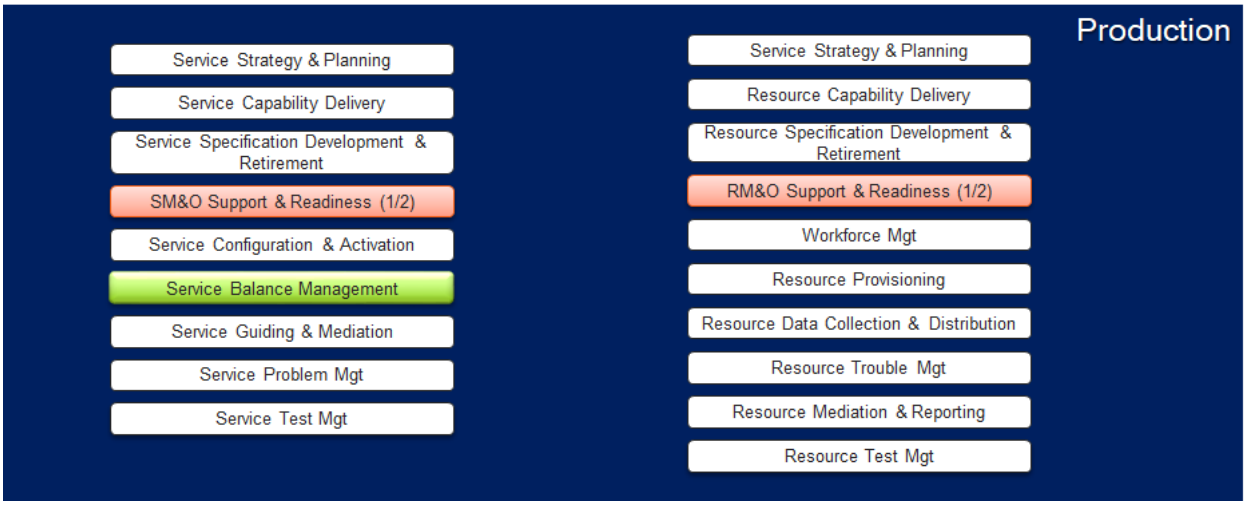

Table 4-9. ***Production Standardized Business Process descriptions (Source reference GB921 R20.5)***

Process Identifier	eTOM Process	Category	Brief Description
1.4.1	Service Strategy & Planning	(2) eTOM Process Type	Enable the development of a strategic view and a multi-year business plan for the enterprise's services and service directions, and

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Process Identifier	eTOM Process	Category	Brief Description
			the parties who will supply the required services.
1.4.2	Service Capability Delivery	(2) eTOM Process Type	Plan and deliver the total capabilities required to deliver changes to service.
1.4.3	Service SpecificationDevelopment & Retirement	(2) eTOM Process Type	Develop and deliver new or enhanced service types.
1.4.4	SM&O Support & Readiness	(2) eTOM Process Type	Manage service infrastructure, ensuring that the appropriate service capacity is available and ready to support the SM&O Fulfillment, Assurance and Billing processes
1.4.5	Service Configuration & Activation	(2) eTOM Process Type	Allocation, implementation, configuration, activation and testing of specific services to meet customer requirements.
1.4.6	Service Problem Management	(2) eTOM Process Type	Respond immediately to customer-affecting service problems or failures in order to minimize their effects on customers, and to invoke the restoration of the service, or provide an alternate service as soon as possible.
1.4.8	Service Guiding & Mediation	(2) eTOM Process Type	Manage usage events by correlating and formatting them into a useful format as well as guiding them to an appropriate service.
1.4.10	Service Test Management	(2) eTOM Process Type	Service Test Management provides the manages the end-to-end execution of a test or test scenario for Services not related to a specific customer's product.
1.5.2	Resource Capability Delivery	(2) eTOM Process Type	Use the capability definition or requirements to deploy new and/or enhanced technologies and associated resources.

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Process Identifier	eTOM Process	Category	Brief Description
1.5.3	Resource SpecificationDevelopment & Retirement	(2) eTOM Process Type	Develop new, or enhance existing technologies and associated resource types, so that new Services can be developed.
1.5.4	RM&O Support & Readiness	(2) eTOM Process Type	Manage resource infrastructure to ensure that appropriate application, computing and network resources are available and ready to support the Fulfillment, Assurance and Billing processes in instantiating and managing resource instances, and for monitoring and reporting on the capabilities and costs of the individual FAB processes.
1.5.5	Workforce Management	(2) eTOM Process Type	Managing the staff performing manual activities along with managing the actual activity being performed. "Note: The current focus of the Manage Workforce processes is field Staff and others managed through work orders, etc. There is opportunity for further study in subsequent releases of eTOM, including: • types and positioning of workforce (field technicians, services representatives, etc.), • other enterprise activities and other management of staff, • decomposition and normalization considering information being acted on.
1.5.6	Resource Provisioning	(2) eTOM Process Type	Allocation, installation, configuration, activation and testing of specific resources to meet the service requirements, or in response to requests from other processes to alleviate specific resource capacity

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Process Identifier	eTOM Process	Category	Brief Description
			shortfalls, availability concerns or failure conditions.
1.5.7	Resource Data Collection & Distribution	(2) eTOM Process Type	Collect and/or distribute management information and data records between resource and service instances and other enterprise processes.
1.5.8	Resource Trouble Management	(2) eTOM Process Type	Responsible for the management of troubles with specific resources.
1.5.10	Resource Mediation & Reporting	(2) eTOM Process Type	Manage resource events by correlating and formatting them into a useful format.
1.5.12	Resource Test Management	(2) eTOM Process Type	Resource Test Management provides the manages the end-to-end execution of a test or test scenario for Resources not specific to a customer's product.
1.X.Y	Service Balance Management	(2) eTOM Process Type	*to be discussed for next versions*

4.5.2. Standard Business Information Objects

Below is SID ABEs for core Production function block

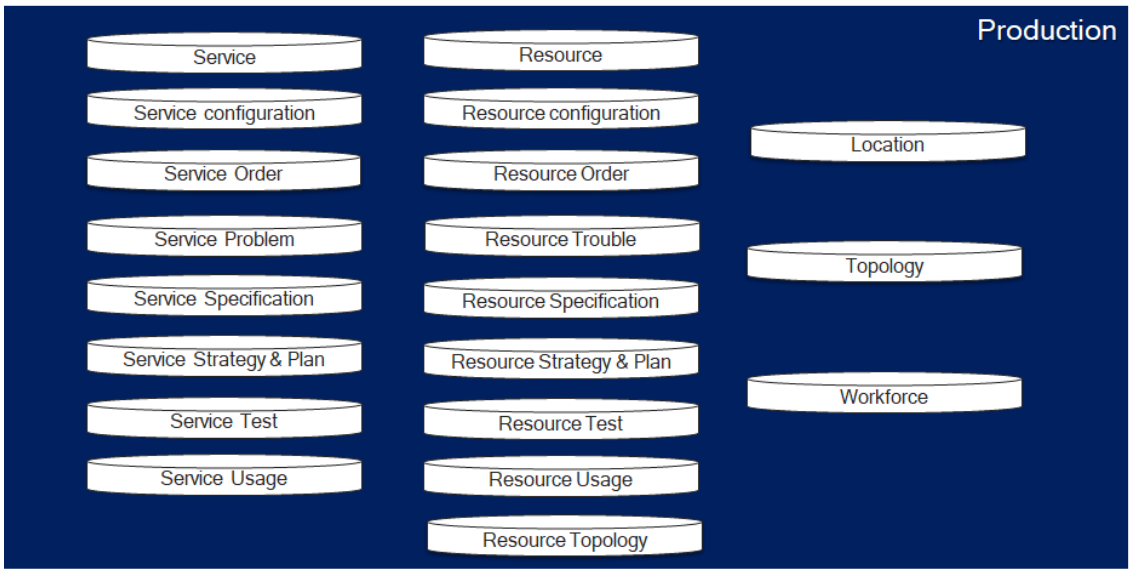

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Table 4-10. ***SID ABEs and their description (extract from GB922 20.5)***

SID Domain	#	SID ABE	ABE Definition
Service	01	Service ABE	Contains entities that are used to represent both customer-facing and resource-facing types of services. Entities in this ABE provide different views to examine, analyze, configure, monitor and repair Services of all types. Entities in this ABE are derived from Service Specification entities.
	02	Service Order ABE	Contains entities that represent a type of Request that decomposes a Customer Order's products into the services associated with a ServiceOrder through which the products are realized.
	03	Service Specification ABE	Contains entities that define the shared characteristics of both types of Service entities. This enables multiple instances to be derived from a single specification entity. In this derivation, each instance will use the characteristics defined in its associated specification. Entities in this ABE focus on adherence to standards, distinguishing features of a Service, dependencies (both physical and logical, as well as on other services), quality, and cost. In general, entities in this ABE enable Services to be bound to Products and run using Resources. Network services are one example of Services that can be built. Additional examples include installation, maintenance, monitoring, and content authentication, authorization, accounting, and auditing functions.
	04	Service Strategy & Plan ABE	Contains entities that are used to address the need for enhanced or new Services, as well as the retirement of existing Services, by the enterprise. These entities have a strong dependency to both entities in the Resource and Product domains. Resulting efforts, such as deciding what Resources to use to host a Service, or what Services are used to support new Product Specifications, are also supported, as are service demand forecasts.
	05	Service Configuration ABE	The definition of how a Service operates or functions in terms of CharacteristicSpecification(s) and related ResourceSpec(s) and ServiceSpec(s) as well as a representation of how a Service operates or functions in terms of characteristics and related Resource(s) and Service(s).

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SID Domain	#	SID ABE	ABE Definition
	06	Service Usage ABE	Collects Service consumption data, and generates Service usage records, for use by other business entities. The entities in this ABE provide physical, logical, and network usage information.
	07	Service Problem ABE	Manages faults, alarms, and outages from a Service point-of-view. This is then correlated to trouble tickets, regardless of whether the cause is physical or logical. Other entities in this ABE are used to direct the recovery from each of these three types of problems. They provide the ability to associate Resource faults and alarms to degradation and outages of Services that run on those Resources. These functions are independent of the Resources and technologies used to build the Service. A third set of entities in this ABE is used to differentiate between customer-reported problems and network-induced problems.
	08	Service Test ABE	A function performed on a service that results in measures being produced that reflect the functioning of the entity under test.
Resource	01	Resource ABE	Contains entities that are used to represent the various aspects of a Resource. This includes four sets of entities that represent: the physical and logical aspects of a Resource; show how to aggregate such resources into aggregate entities that have physical and logical characteristics and behavior; and show how to represent networks, sub-networks, network components, and other related aspects of a network.
	02	Resource Order ABE	Contains entities related to the business interaction orders.
	03	Resource Specification ABE	Contains entities that define the shared characteristics and behavior of each of the four types of Resource entities. This enables multiple instances to be derived from a single specification entity. In this derivation, each instance will use the shared characteristics and behavior defined in its associated template.
	04	Resource Topology ABE	Contains entities that define physical, logical, and network topological information. This information is critical for assessing the current state of the network, as well as providing information on how to fix problems, tune performance, and in general work with the network (both as a whole and with its components). Each of these topological views provides its own physical, logical, or network related

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SID Domain	#	SID ABE	ABE Definition
			information that can be used to manage one or more layers in a layered network.
	05	Resource Configuration ABE	The definition of how a Resource operates or functions in terms of CharacteristicSpecification(s) and related ResourceSpec(s), as well as a representation of how a Resource operates or functions in terms of characteristics and related Resource(s).
	06	Resource Usage ABE	Collects Resource consumption data, and generates Resource usage records, for use by other business entities. The entities in this ABE provide physical, logical, and network usage information.
	07	Resource Strategy & Plan ABE	Is used to plan networks and resource elements both initially and for growth. It will coordinate both logical and physical resource growth. Inputs are budgets from business sources, service forecasts, current and projected network utilization, new technologies, and retiring technologies. It handles the lifecycle (installation, modification, removal, and retirement) for both logical and physical resources.
	08	Resource Trouble ABE	Manages problems found in allocated resource instances, regardless of whether the problem is physical or logical. Entities in this ABE detect these problems, act to determine their root cause, resolve these problems and maintain a history of the activities involved in diagnosing and solving the problem. Detecting problems can be done via software (e.g., responding to an alarm) and/or by hardware (e.g., a measurement or probe) and/or manually (e.g., visual inspection). This includes tracking, reporting, assigning people to fix the problem, testing and verification, and overall administration of repair activities.
	09	Resource Test ABE	Contains entities that are used to test PhysicalResources, LogicalResources and CompoundResources. These entities are usually invoked during installation, as a part of trouble diagnosis or after trouble repair has been completed.
Enterprise	01	Workforce ABE	Represents dispatched Field Force Management - human and other field resource with their roles, skills, calendars and other characteristics. Also, models work orders, catalog of work descriptions, various schedules, as well as reservation and assignment of a technician to a task
Common	01	Topology ABE	The Topology ABE contains entities that are used to represent topological concepts that can be used to model a

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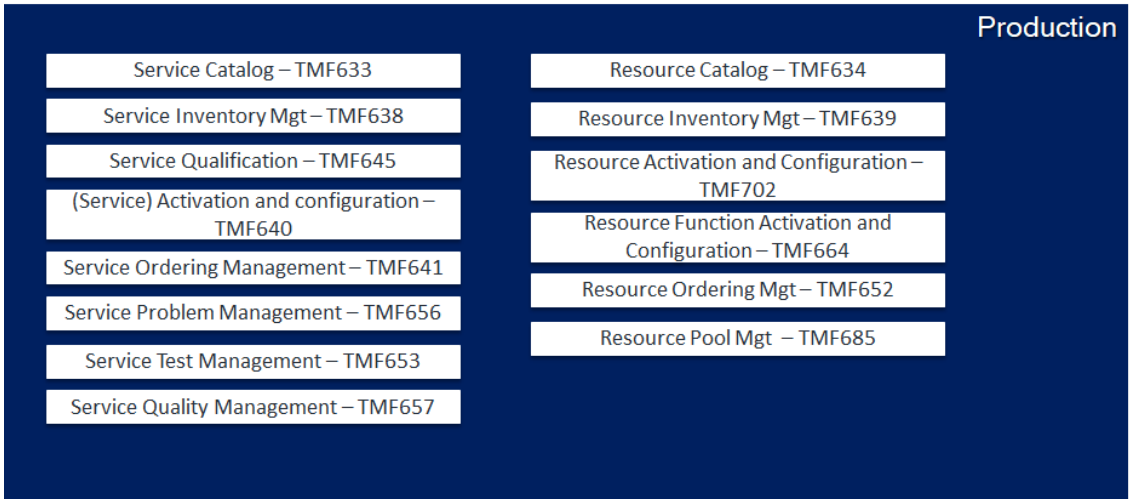
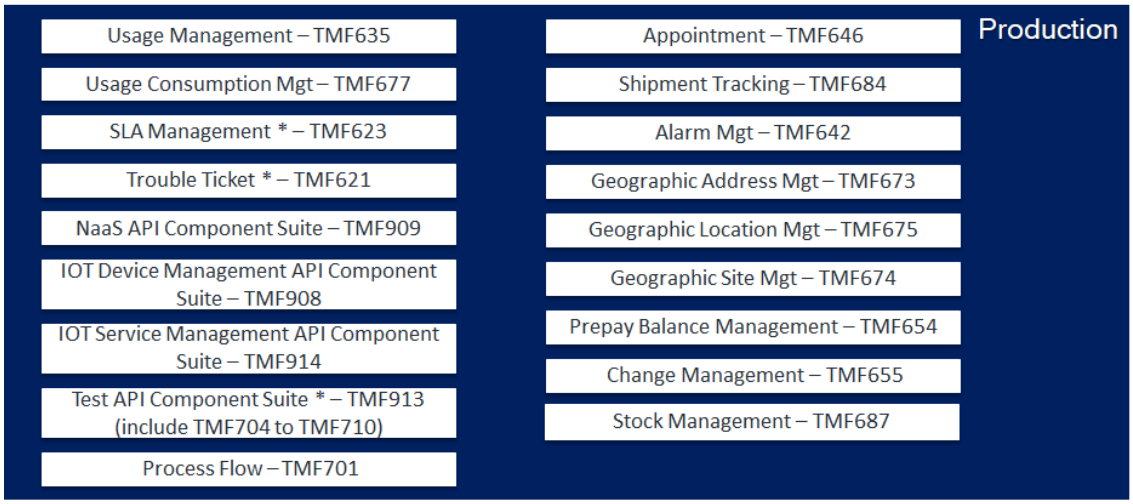
SID Domain	#	SID ABE	ABE Definition
			large variety of topological relations between entities ranging from dependencies to connectivity.
	02	Location ABE	The Location ABE represents the site or position of something, such as a customer's address, the site of equipment where there is a fault and where is the nearest person who could repair the equipment, and so forth. Locations can take the form of coordinates and/or addresses and/or physical representations

4.5.3. Standardized Interfaces





* Shared with Core Commerce for Product level – further study required

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Table 4-11. ***Standard Interfaces in the Production block***

#	Interface Name	Purpose / Descriptions	Specification / Reference
Service related:			
01	Service Catalog API		Open API TMF633
02	Service Inventory Management API		Open API TMF638
03	Activation and configuration API		Open API TMF640
04	Service Ordering Management API		Open API TMF641
05	Service Qualification API		Open API TMF645
06	Service Test Management API		Open API TMF653
07	Service Problem Management API		Open API TMF656
08	Service Quality Management API		Open API TMF657
Resource related:			
01	Resource Catalog Management API		Open API TMF634
02	Resource Inventory Management API		Open API TMF639
03	Resource Ordering Management API		Open API TMF652
04	Resource Function Activation and Configuration API		Open API TMF664
05	Resource Pool Management API		Open API TMF685
06	Resource Activation and Configuration API		Open API TMF702
Common API's for Service and Resource:			
01	Trouble Ticket API		Open API TMF621
02	SLA Management API		Open API TMF623
03	Usage Management API		Open API TMF635
04	Alarm Management API		Open API TMF642
05	Appointment API		Open API TMF646
06	Prepay Balance Management API		Open API TMF654
07	Change Management API		Open API TMF655
08	Geographic Address Management API		Open API TMF673

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#	Interface Name	Purpose / Descriptions	Specification / Reference
09	Geographic Site Management API		Open API TMF674
10	Geographic Location Management API		Open API TMF675
11	Usage Consumption Management API		Open API TMF677
12	Shipment Tracking API		Open API TMF684
13	Stock Management		Open API TMF687
14	Process Flow API		Open API TMF701
15	IOT Device Management API Component Suite		Open API Suite TMF908
16	NaaS API Component Suite		Open API Suite TMF909
17	Test API Component Suite		Open API Suite TMF913
18	IOT Service Management API Component Suite		Open API Suite TMF914

4.6. Intelligence Management

Definition and Scope

Analytical processes use operational data (reference data) and events produced by operational processes or captured from the environment during an interaction with the enterprise (for instance, events related to the activity of a user on a portal, events generated by operating connected object or devices).

Analytical processes use operational data along with events received from operational processes to produce analyses, correlations and aggregations that yield a representation of operational reality. Analytical processes encompass activity or budget follow-up, as well as revenue assurance or fraud detection, based on traffic analysis.

Analytical processes can use very large amounts of data and in turn produce their own data (e.g., all kinds of KPI) for which they are entirely responsible and which they may make available to other systems (either for display, or as input data for further processing).

Therefore, analytical processes are distinct from operational processes but interact with them through event exchanges and data sharing. Big and Fast data capabilities are expected to further develop such interactions and enhance operational processes.

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4.6.1. Business Processes in the Intelligence Management Block

Based on the figure below, eTOM level 2 are identified as "split into 2" when they manage SID ABEs that are mapped to different ODA functional blocks.

The Intelligence Management process descriptions (Source GB921 R20.5)

Table 4-12. ***Intelligence Management eTOM Process descriptions***

Process Identifier	eTOM Process	Category	Brief Description
1.1.7	Market Sales Support & Readiness	(2) eTOM Process Type	Market Sales Support & Readiness processes ensure the support capability is in place to allow the CRM Fulfillment, Assurance and Billing processes to operate effectively.
1.1.12	Market Performance Management	(2) eTOM Process Type	Market Performance Involves managing, tracking, monitoring, analyzing, improving and reporting on the performance of market key performance indicators
1.1.13	Sales Performance Management	(2) eTOM Process Type	Sales Performance Management managing, tracking, monitoring, analyzing, improving and reporting on the performance of specific sales key performance indicators that can be defined for certain levels, such as sales channels, organizations, and so forth.
1.1.15	Marketing Campaign Management	(2) eTOM Process Type	Develop a marketing campaign through a coordinated series of steps that can include promotion of a product through different mediums (television, radio, print, online) using a variety of

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Process Identifier	eTOM Process	Category	Brief Description
			different types of advertisements to reach the market, customers and channels.
1.2.4	Product Support	(2) eTOM Process Type	Product Support processes ensure the support capability is in place to allow the CRM Fulfillment, Assurance and Billing processes to operate effectively.
1.2.6	Product Performance Management	(2) eTOM Process Type	Manage, track, monitor, analyze, improve, and report on the performance of specific products.
1.2.13	Product Test Quality Analysis	(2) eTOM Process Type	Product Test Quality Analysis manages a quality Analysis on Tests to continually tweak and improve it.
1.3.1	Customer Support	(2) eTOM Process Type	Customer Support processes ensure the support capability is in place to allow the CRM Fulfillment, Assurance and Billing processes to operate effectively.
1.3.2	Customer Experience Management	(2) eTOM Process Type	Perform customer lifecycle mapping, assess customer experience maturity, as well as analyze customer experience metrics.
1.4.4	SM&O Support and Readiness	(2) eTOM Process Type	Manage service infrastructure, ensuring that the appropriate service capacity is available and ready to support the SM&O Fulfillment, Assurance and Billing processes
1.4.7	Service Quality Management	(2) eTOM Process Type	Managing, tracking, monitoring, analyzing, improving and reporting on the performance of specific services
1.4.11	Service Test Quality Analysis	(2) eTOM Process Type	Service Test Quality Analysis manages a quality Analysis on Tests to continually tweak and improve it.
1.5.4	RM & O Support and Readiness	(2) eTOM Process Type	Manage resource infrastructure to ensure that appropriate application, computing and network resources are available and ready to support the Fulfillment, Assurance and Billing processes in instantiating and managing resource instances, and for monitoring and reporting on the capabilities and costs of the individual FAB processes. *(Ongoing Study)*

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Process Identifier	eTOM Process	Category	Brief Description
1.5.9	Resource Performance Management	(2) eTOM Process Type	Managing, tracking, monitoring, analyzing, controlling and reporting on the performance of specific resources
1.5.13	Resource Test Quality Analysis	(2) eTOM Process Type	Resource Test Quality Analysis manages a quality Analysis on Tests to continually tweak and improve it.
1.6.6	Party Support	(2) eTOM Process Type	Party Support processes are responsible for ensuring that all necessary facilities related to the interaction with other parties are ready and functioning. Moreover, these processes are responsible for the resolution of problems related to these facilities. *(Ongoing Study)*
1.6.11	Party Performance Management	(2) eTOM Process Type	Track, monitor and report on performance to ensure that the interactions are in accordance with the agreed commercial arrangements between the service provider and another party.

4.6.2. Standard Business Information Objects

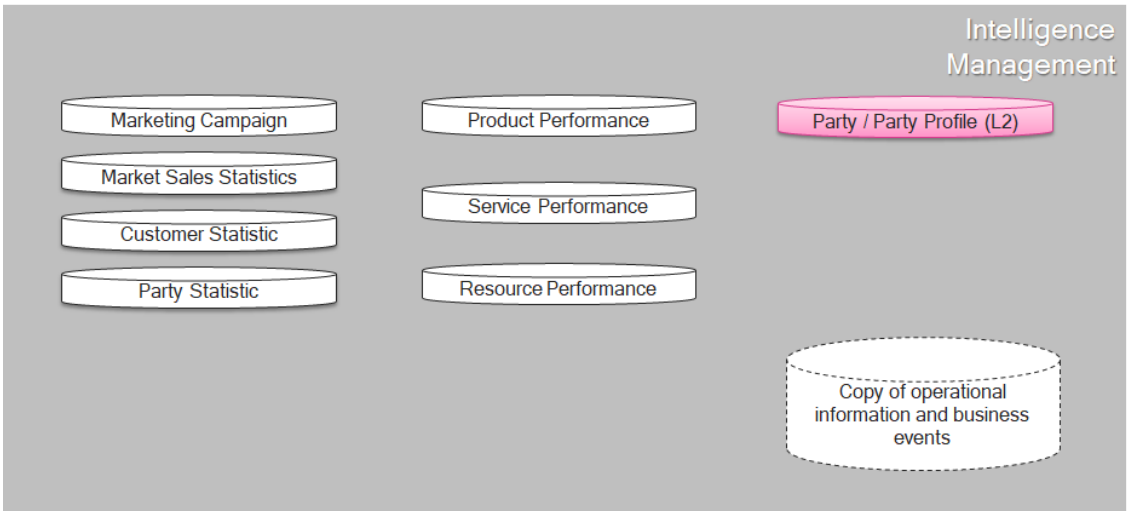

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Table 4-13. ***SID ABEs and their description (extract from GB922 20.5)***

SID Domain		#	ABE	ABE Definition
Market / Sales		01	Marketing Campaign ABE	Supports marketing new or existing product offerings to identified target markets. For example, the launch of a pre-paid product with multiple promotions across distribution channels, market segments and so forth; a new campaign for an existing product; a re-launch of a campaign for an existing product.
		02	Market Sales Statistics ABE	Maintains sales forecasts, new service requirements, customer needs, and customer education, as well as calculating key performance indicators about Sales & Marketing revenue and sales channel performance.
Customer		01	Customer Statistic ABE	Represents the analysis of customer usage patterns, customer profitability statistics and churn and retention statistics.
Product		01	Product Performance ABE	Handles product performance goals, the results of end-to-end product performance assessments, and the comparison of assessments against goals. The results may include the identification of potential capacity issues.
Service		01	Service Performance ABE	Collects, correlates, consolidates, and validates various performance statistics and other operational characteristics of customer and resource facing service entities. It provides a set of entities that can monitor and report on performance. Each of these entities also conducts network performance assessment against planned goals, performs various aspects of trend analysis, including error rate and cause analysis and Service degradation. Entities in this ABE also manage the traffic generated by a Service, as well as traffic trend analysis. This is important for newer technologies that separate data, control and management functions for a given Service.
Resource		01	Resource Performance ABE	Collects, correlates, consolidates, and validates various performance statistics and other operational characteristics of Resource entities. It provides a set of entities that can monitor and report on performance. The entities in this ABE provide physical, logical, and performance information. Each of these entities also conducts network performance assessment against planned goals, performs various aspects of trend

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SID Domain		#	ABE	ABE Definition
				analysis, including error rate and cause analysis and Resource degradation. Entities in this ABE also manage traffic in a Resource. This includes statistics defining Resource loading, and traffic trend analysis.
Common		01	Party / Party Profile (L2)	The characteristics used to group Parties that typify MarketSegments and for the formulation and targeting of MarketingCampaigns. A Party ProfileType is used to categorize one or more PartyRoleSpecifications. A PartyProfileType can be defined for one or more GeographicAreas. The profile can target one or more ProductOfferings and one or more MarketSegments. PartyProfileTypes can be based on (or defined by) DemographicCharacteristics.
Business Partner		01	Party Statistic ABE	Represents the analysis of party usage patterns, profitability statistics

4.6.3. Intelligence Management Block functionalities

A list of Intelligence Management functionalities identified as part of the ongoing study. The functionalities currently found to be provided by Intelligence Management are captured below.

Table 4-14. Functionalities covered by the Intelligence Management block

#	Functionality	Description / Purpose
01	Insight Management	Insight Management integrates data in the Engagement Management Block, Party Management Block, Core Commerce Management Block and Production Block, to find new relationships and patterns by analyzing historical data, assessing current situation, applying business rules, predicting outcomes, and proposing the next best action.
02	Autonomic Manager Elements	The Autonomic Manager Element's function or Decision Making Element (DE) is to drive control-loops meant to configure and adapt (i.e., regulate) the behavior or state of an ODA resource. Autonomic Management can be implemented to run as an “engine”. (Reference - Generic Autonomic Network Architecture "GANA".http://www.etsi.org/deliver/etsi_ts/103100_103199/10319502/01.01.01_60/ts_10319502v010101p.pdf).
03	Algorithm Ma nagement	This function provides the environment to develop and deploy patterns or rules or outcome based "recipes" for business and operational actions performed by Parties or the Autonomic Manager. It gives developers the ability to edit, compose, integrate, package, train and deploy AI-based microservices. (Reference - https://docs.acumos.org/en/latest/)

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#	Functionality	Description / Purpose
04	Analytic Models	Analytics models provide input for Insight Management, functions based on analytic processes (Reference GB979D Big Data Analytics Big Data Repository). Analytic models do not change any operational data but provide functionality to discover, acquire, translate and communicate data and knowledge (synthesized from raw data), relevant to making decisions, feeding decisions or for autonomics. A Cognitive Autonomic Manager (in ETSI GANA Model terms, a Cognitive Decision Element (DE)) as a Deployable AI Model, employs in its operations.
05	Data Lake	The Data Lake function, acquires, manages or prepares data for use by ODA and the enterprise. It includes a data collector function and can translate data from multiple sources in order to create federated information or data models (Data processing and Data storage shall be separated).
06	Knowledge Plane	Knowledge Plane references include ETSI GANA (Generic Autonomic Network Architecture) Knowledge Plane specified in ETSI TS 103 195-2 but not limited to it.
07	Governance Function (Dashboard)	It is an instrument that allows the CSP defining Business Policies, Objective /Goal and Intent specifications to be enforced and executed to ensure an optimal operation and achievement of desired objectives and services while keeping control of activation / deactivation of deployed DEs (AI Models), conflicts resolution that may happen and all associated accountability and responsibility aspects
08	Training Data Repository	Various Datasets (e.g., Data extracted from an ODA Data Lake) must be considered in populating the Data Training Repository in order to produce clean and accurate Data and also synthesize knowledge from the data and make the data/knowledge available for AI Model (e.g., Cognitive Autonomic Manager Element/ Cognitive Decision Element (DE)) training and testing processes
09	Testing and Evaluation	Testing and Evaluation System/Component for Testing and Evaluation of the Performance and Quality of “Decision-Making capability” of Cognitive Autonomic Manager Element (DE) as an AI Model, against the expected /predicted targets (prescribed by the CSP) in the CSP’s Testing Environment before declaring and certifying the AI Model (DE) as “operation ready” and “deployable AI Model (DE)” for injection into a Network Element/Function (NE/NF) in the ODA Production or any other ODA Functional Bock or for operationalizing it in the ODA Knowledge Plane (KP) if it is a Knowledge Plane Level DE.

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4.6.4. Standardized Interfaces

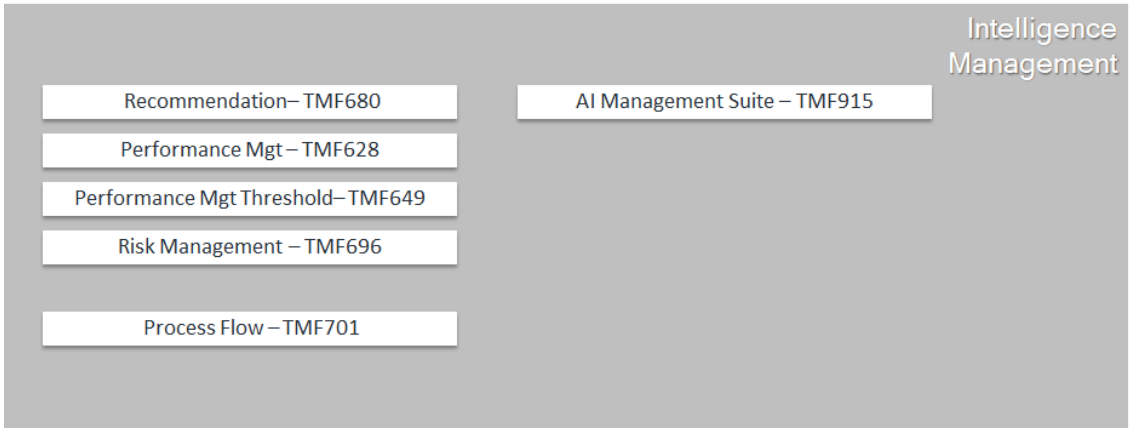

Table 4-15. ***Standard Interfaces in the Intelligence Management block***

#	Interface Name	Purpose/Description	Specification / Reference
1	Performance Management API	A TMF Defined Open API	Open API TMF628
2	Performance Management Threshold API	A TMF Defined Open API	Open API TMF649
3	Recommendation API	A TMF Defined Open API	Open API TMF680
4	Risk Management	A TMF Defined Open API	Open API TMF696
5	Process Flow API	A TMF Defined Open API	Open API TMF701
6	AI Management Suite	A TMF Defined Open API Suite	Open API Suite TMF915

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5. Using the Functional Architecture

5.1. Reference User Stories

2 types of user stories and use cases are available :

- set of simple business use cases to illustrate how ODA can easily solve current issues
- dynamic illustration of ODA/Open APIs call flow use cases to illustrate the Order to Cash global process

5.1.1. IG1197 ODA User Stories and Use Cases

5.1.2. IG1228 ODA/APIs Call Flow Use Cases

5.2. Function level mappings

As the Functional Framework does not yet permit to easily identify functions supporting a business activity (eTOM level 2 or 3) and SID ABEs or BEs produced, we only have an example of functions mapping in IG1167 ODA Functional Architecture Exploratory report - Section 2.

5.3. Creating an ODA Component

5.3.1. IG1171 ODA Component Definition

5.4. Implementation Scenarios

5.4.1. IG1167 ODA Functional Architecture

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6. Evolution

New opportunities, new concepts and new technology will continue to impact on the capability of service provider technology management and operational systems. The fast pace of Open source innovation, cross-industry collaboration and inside-out innovation by members will help advance the granularity of ODA specifications.

The ODA Functional Architecture working group will evaluate, on a continuing basis, impact of new technologies, concepts and technology and identify enhancements through the Introductory Guide IG1167. Following active member collaboration and contributions, ongoing reviews and joint approvals will drive the evolution of this standard.

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7. Administrative Appendix

7.1. Terminology, Abbreviations and Definitions

Reference TMF071 ODA Terminology R19.0.1

7.2. References

7.2.1. Core References

- Open Digital Architecture Whitepaper - https://www.tmforum.org/resources/whitepapers/open-digital-architecture/
- Open Digital Architecture Vision - IG1166 Open Digital Architecture Vision R18.0.0
- Open Digital Architecture Concepts & Principles - GB998 Open Digital Architecture (ODA) Concepts & Principles R18.0.0
- Open Digital Architecture Introductory Guide - IG1167
- OSS of the Future - https://www.tmforum.org/resources/whitepapers/oss-of-the-future/

7.2.2. Other Related References

- IG1186 ODA Governance and Security Vision R19.0.1
- IG1187 ODA Enterprise Risk Assessment R19.0.1
- IG1176 TOSCA Guide for Model-Driven Automation R19.0.0
- GB999 ODA Production Implementation Guidelines R19.0.0
- IG1171 ODA Component Definition R19.0.0
- IG1130F SDN/NFV impacts on TAM Knowledge Management R19.0
- IG1157 Digital Platform Reference Architecture Concepts and Principles R19.0.1

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8. Document History

8.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
1.0.0	02-Oct-2020	Emmanuel A. Otchere Sylvie Demarest Alan Pope	Final edits prior to publication
1.1.0	29-Jan-2021	Alan Pope	Final edits prior to publication
Production	26-Mar-2021	Adrienne Walcott	Updated to reflect TM Forum Approved Status

8.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	02-Oct-2020	Alan Pope	Initial version
Production	24-Nov-2020	Adrienne Walcott	Updated to reflect TM Forum Approved Status
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- Dave Milham, TM Forum

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